ESM

PLANT AUDIT REPORT

Industrial Operations, Facility & Quality Compliance Assessment

DRAFT REPORT

AUDIT DETAILS & FACILITY INFORMATION

Audit Reference:	**AUD-000001**	Company:	**Neptune Plast Machinery Ltd.**
Audit Type:	Comprehensive Plant Audit	Plant Facility:	Vadodara Emitting Pipe Facility (Vadodara)
Audit Date:	13/7/2017	Auditor / Employee	Aarav Sharma (Lead Auditor) (ESM-EMP-001)

EXECUTIVE AUDIT SUMMARY

136	65%	78	0	10
TOTAL QUESTIONS	COMPLETION %	COMPLIANT	NON-COMPLIANT	OBSERVATIONS

CRITICAL & HIGH ACTION REGISTER

ID	Finding Description	Severity	Corrective Action	Target / Owner
Q006	This factory was started in 2003 and currently has...	**Medium**	Review operational controls and establish pre...	Plant Operations Manager / 22/11/2026

SECTION A: Equipment & Facilities

Q006 OBSERVATION

Attributes of key major manufacturing equipment:

Response: This factory was started in 2003 and currently has 16 operating lines making 12 & 16 mm emitting pipes and lateral pipes . Also has assembly for dripp

Q007 COMPLIANT

Mixers (# of mixers, manufacturer, model, capacity, age, general appearance) raw material and accessories

Response: Updated high precision extruder operational parameters test.

Observation: Auditor verified during test run.

Q008 COMPLIANT

Number of extrusion lines

Response: 16 lines, unit 1 - 10 lines and Unit 2 - 6 lines . Unit 1 was started with 3 lines in 2003 and subsequently expanded to 10 lines . Unit 2 was set up i

Q009 COMPLIANT

Main lines equipment (manufacturer, model, capacity, age)

Response: Most of the extruders are KABRA INDIA make with outputs of 200 to 350 kg/hr . The downstream equipment's including the auto coilers are supplied from

Q010 PENDING

# of SKUs manufactured per line

Response: No response recorded

Q011 COMPLIANT

General lines appearance

Response: Well maintained and upgraded continously , break down in under 1%

Q012 PENDING

Approximate equipment value

Response: No response recorded

Q014 COMPLIANT

Owned or leased Facility?

Response: Premises is on lease of 99 years

Q015 PENDING

Any current government incentive for operating in the location?

Response: No response recorded

Q016 PENDING

If owned, approximate land value?, any current government incentive for operating in the location?

Response: No response recorded

Q017 COMPLIANT

Is there a warehouse for raw material? Capacity? Location?

Response: Warehouse is around 30,000 square feet for accessories/bought out items/RM etc and is within the plant site

Q018 COMPLIANT

Is there a warehouse for finished products? Capacity? Location?

Response: FG storage is in open area

Q020 COMPLIANT

Observe condition, note recent capital improvements or maintenance needs identified to:

Response: Capital investment of $ 320 K in 2017 for upgrading the downstream machines and controls, to make line flexible for size changeovers etc. With this in

Q021 PENDING

Exterior

Response: No response recorded

Q022 COMPLIANT

Building exterior

Response: Pre fabricated building

Q023 COMPLIANT

Roof

Response: Metal roofing

Q024 COMPLIANT

Parking lot (including lighting)

Response: Yes

Q025 COMPLIANT

Landscaping

Response: Yes

Q026 COMPLIANT

Loading docks

Response: 7 numbers

Q027 COMPLIANT

Neighboring properties

Response: Sabic Research centre, Thermax, Du pont, Penwalt

Q028 PENDING

Interior

Response: No response recorded

Q029 COMPLIANT
Walls
Response: Brick walls upto 5 meter , after that prefabricated

Q030 COMPLIANT
Ceilings
Response: metal roofing

Q031 COMPLIANT
Floors
Response: Tremixing VDF Floors

Q032 COMPLIANT
Office space, conference rooms
Response: Conference room for 10

Q033 COMPLIANT
Restrooms
Response: Adequate

Q034 COMPLIANT
Cafeteria (food service and/or room for self-service?)
Response: Canteen facility for all employees

Q036 OBSERVATION
Own well or water from city network?
Response: water requirement is 1500 lits per day and is met by city network, no borewell

Q037 COMPLIANT
Sewage (connection to city drainage or own sewage treatment plant?)
Response: Own STP plant and the treated water is used for gardening

Q039 PENDING
Square footage breakdown (some information already available to be captured before visits)
Response: No response recorded

Q040 COMPLIANT
Total site area
Response: 33174 Square meter

Q041 COMPLIANT
Total roofed area
Response: 60% is built up area

Q042 PENDING
Plant area
Response: No response recorded

Q043 COMPLIANT
Manufacturing
Response: 5000 Square meter

Q044 COMPLIANT
Distribution and storage (receiving, shipping)
Response: 3000 Square meter

Q045 PENDING
Support areas (office space, conference rooms, restrooms, lunchrooms, break rooms, etc.)
Response: No response recorded

Q046 COMPLIANT

Parking lots area

Response: Vehicle parking is available for employees vehicles

Q047 COMPLIANT

Opportunities for plant expansion (either inside building footprint, on site, or on adjacent sites)

Response: Unit 2 has space to accommodate another 4 extrusion lines

SECTION B: Plant Management / Leadership

Q049 COMPLIANT

What is the tenure/level of experience of top management (years)?

Response: The Plant Manager and SCM Head are well experinced with more than 20 years of relevant experience and are with company for more than 15 years.

Q050 COMPLIANT

What is the turnover rate for plant management in last 2 years? What management retention programs are in place? Is there a formal succession plan?

Response: Average years in the company is 9-11 years

Q051 PENDING

Type of training for plant management / leadership?

Response: No response recorded

SECTION C: Plant Labor

Q053 COMPLIANT

Total plant headcount? Direct? Indirect?

Response: Company employees in Manufacturing - 77 (Production/Quality/Maint), Contract labours - 134 for operations like coiling/packing/ shifting/grinding etc.

Q054 OBSERVATION

Is the plant union or non-union?

Response: NO Union

Q055 COMPLIANT

If union, is there an affiliation to a workers Organization?

Response: N/A

Q056 COMPLIANT

If union, what´s the union structure (e.g., one delegate vs. several commisioners)?

Response: N/A

Q057 COMPLIANT

Describe union labor contracts and how they are managed

Response: N/A

N058 COMPLIANT

What is the collective agreement value?

Response: N/A

Q059 COMPLIANT

Are the employees hired directly by the company or outsourced?

Response: The contract workers are hired thru a contractor company , the wage negotiation is done with the contractor once in 3 years. Wages are higher than min

Q060 COMPLIANT

What is the average employee demographics (age range, gender, education level)?

Response: Average employee age is 28-31 years

Q061 COMPLIANT

How does employee pay compare to other jobs in the local area?

Response: It is in line with the industry practice in the area

Q062 COMPLIANT

What is the turnover rate for plant labor over the last 2 years?

Response: Hardly any turnover

Q063 PENDING

What is the absenteeism rate for plant labor over the last 2 years?

Response: No response recorded

Q064 PENDING

Do you conduct employee satisfaction or working environment surveys? If so, how often? when was the most recent and what were the results?

Response: No response recorded

Q065 PENDING

What is the onboarding process for new employees (salaried and hourly)?

Response: No response recorded

Q066 COMPLIANT

Is there a comprehensive training and certification program in place for production associates? What type of training / certifications?

Response: Focus is on multiskilling training programme

Q067 COMPLIANT

Do you have some performance based incentive programs and recognition program to motivate employees engaged in manufacturing operations? If yes, how does the incentive program work (e.g., dimensions considered, type of incentives)?

Response: R & R (REWARDS & RECOGNITION) programme is in place for Kaizens and Star performers

SECTION D: Capacity Utilization

Q069 OBSERVATION

What are the facility and process bottleneck areas (e.g., machines, people, material, storage, space)? How are they being mitigated (if at all)?

Response: Apparently there does not seem any major bottleneck, the plant is working on Lean manufacturing practises in each area and showing improvements by val

Q070 COMPLIANT

What are the key metrics for capacity and productivity (e.g., ton/employee/month)? How are they derived?

Response: The key metrics which are mapped are like OEE, scrap %, meters production, units/meter etc . These are measured and monitored on real time basis with

Q071 PENDING

How has the transformation cost changed over time? Why?

Response: No response recorded

Q072 PENDING

What efforts have been made over time to increase performance?

Response: No response recorded

Q073 PENDING

Total Volume produced annually? By product type (80/20)?

Response: No response recorded

Q074 COMPLIANT

How much additional volume could be added to this plant before additional capacity is needed?

Response: 10-15%, (862 mm to 988 mm)

Q075 PENDING

Raw material Cost- Rs/Kg

Response: No response recorded

Q076 PENDING

Transformation Cost- Rs/Kg

Response: No response recorded

Q077 COMPLIANT

Energy consumption - Units/Kg

Response: Energy units is 22 KWH/meter

Observation: Alternate Reference: Energy consumption (kW / ton)?

Q078 PENDING

Utilization

Response: No response recorded

Q079 OBSERVATION

How is availability determined?

Response: Calender hours less no plan

Q080 COMPLIANT

What is the equipment that presents the most problems?

Response: Downstream equipments like coilers

Q081 COMPLIANT

How product and size changeovers affect utilization?

Response: 70% of the volume is in 16 mm pipes, hence it helps in continous running of 7-8 machines with little size changeovers and reduced scrap levels. For ba

Q082 PENDING

What are the biggest sources of utilization loss?

Response: No response recorded

Q083 COMPLIANT

Are there scheduled down times (e.g. cleaning, maintenance)?

Response: Yes preventive maintenance schedules are made and implemented for all lines. The braekdown is under 1%

Q084 PENDING

Throughput

Response: No response recorded

Q085 COMPLIANT

How have you established maximum speed or throughput rate?

Response: Based on the maximum line speed of each line in a particular size, means the line speed of 12 mm will be higher than 16 mm pipes. The average line spe

Q086 COMPLIANT

Are there process engineering teams to determine maximum demonstrated speed at different lines?

Response: Yes there is lean team working on the various productivity improvement projects including value engineering

Q087 PENDING

Acceptance

Response: No response recorded

Q088 COMPLIANT

How do you measure quality (metrics, procedures, tests, laboratory)?

Response: As per IS quality plan

Q089 COMPLIANT

How materials efficiency / yield is determined and tracked (e.g., scrap, extraweight)?

Response: The scrap/OEE/weight per meter etc are tracked real time thru automation on the machines (thru PLC) and displayed real time

Q090 COMPLIANT

How do you measure scrap, extraweight?

Response: As above

SECTION E: Plant Maintenance

Q092 COMPLIANT

Describe the plant preventive maintenance program

Response: Plant has well established preventive maintenance plan

Q093 COMPLIANT

Do you use predictive maintenance?

Response: No

Q094 COMPLIANT

What percentage of total maintenance down time is corrective vs. other types of maintenance (predictive and preventive)?

Response: The break down is under 1%

Q095 COMPLIANT

How is maintenance scheduled?

Response: Once in a month and during size changes

Q096 COMPLIANT

How is preventive maintenance managed? In house or by a third party?

Response: Mostly inhouse

Q097 PENDING

Are there deferred maintenance items that need to be addressed but have not been due to cost?

Response: No response recorded

Q098 PENDING

and role of people in the maintenance department?

Response: No response recorded

SECTION F: Quality

Q100 COMPLIANT

What initiatives are currently underway in the plant to improve quality?

Response: They calculate cost of poor quality on monthly basis and it is showing downward trend on yearly basis. In addition they calculate line wise CP/CPK for

Q101 COMPLIANT

Describe process for inspecting incoming raw materials and approving shipments of finished goods

Response: All incoming material is approved as per test reports and the finished goods are checked /approved as per IS standards

Q102 OBSERVATION

Is there a quality management system? How does it work? Is there a plan and process for tests (defined tests, frequency)?

Response: Yes they have QMS in place with test plans for incoming, in process and final inspection. They do not have third party inspection , the FG inspection

Q103 COMPLIANT

Are they quality certified? If so, how were they certified?

Response: Yes ISO 9001 - 2008 version, certified by International Certification Services with validity upto April 2018

Q104 COMPLIANT

Is the laboratory certified? Internal or external laboratory?

Response: The lab is set up as per requirement of Indian standards and some QA equipment is as per Natafim Israel bencmarking including ESCR

Q105 COMPLIANT

Are products certified?

Response: Yes

Q106 PENDING

and role of people in the quality department?

Response: No response recorded

SECTION G: Environmental Health and Safety

Q108 OBSERVATION

What are the key metrics tracked for plant safety related to frequency and severity (e.g., recordable incident rate, lost time incident rate)?

Response: They are tracking the TRIR/LTIR/SIR and other safety metrics. In 2016 and 2017 there is no recordable accident though there are cases of first aids

Q109 COMPLIANT

What/when were the last major safety incidents / violations in the plant? How were they resolved?

Response: NIL

Q110 OBSERVATION

What initiatives are currently underway in the plant to improve safety?

Response: Focussed improvement plans for like safety audits, safety committee meetings, safety observation drives, daily shop floor meetings, electrical audits

Q111 OBSERVATION

Is there a safety management system? How does it work?

Response: They are certified with ISO 14001 but not with OHSAS 18001

Q112 COMPLIANT

Are they safety certified? If so, how were they certified?

Response: They are certified with ISO 14001 - 2004 VERSION and validity is upto December 2018

Q113 COMPLIANT

Does the plant have a Safety and Health Committee?

Response: Yes they have safety committee which meets once in a month, they also have daily shop floor meetings where safety matters are discussed . There is des

Q114 PENDING

Describe current safety programs (e.g. HACCP - Hazard Analysis and Critical Control Points, ISO 22000:2005)?

Response: No response recorded

Q115 OBSERVATION

Are there any past or threatened/pending legal or environmental issues?

Response: The factory is situated in green zone belt with no pending voilation of Pollution and other regulations

Q116 PENDING

and role of people in the environmental, health and safety department?

Response: No response recorded

SECTION H: Continuous Improvement

Q118 COMPLIANT

Describe any corporate or plant specific improvement initiatives / projects (e.g., for cost reduction), objectives and results

Response: The key objectives and Indicators are 1) zero accidents 2) Cost of quality , reduction targets each year 3) Defined value engineering and productivity

Q119 COMPLIANT

Are your shop floor employees involved in driving process and product improvement? Are there incentives to foster involvement?

Response: Yes the shop floor employees are involved in idea generation and improvements. The Kaizen programme is in place . Yes the members are suitably reawrde

Q120 COMPLIANT

Are your shop floor workers empowered to make recommendations on changes to manufacturing processes, policies and procedures?

Response: The shop floor workers are empowered

Q121 COMPLIANT

Have you documented all the manufacturing process? Do you review them regularly for improvement? What is the process for carrying out change in a existing process? Do you document the changes?

Response: N/A

Q122 COMPLIANT

What systems do you use to perform planning, functions for plants/DCs/materials (e.g., any ERP)? How often are the system used? What modules / functionality is implemented? Do these systems interface with other systems?

Response: SAP ECC is implented along with Planning module and MM. The bar coding of products including that of ware house accessories etc in place. The add on

Q123 COMPLIANT

Describe any continuous improvement methodologies in place and what metrics apply (e.g. Lean / Six Sigma)

Response: Lean journey started in manufacturing and SCM in 2014. Lean deployment plan is made at the start of the year aong with deliverables, weekly meeting fo

Q124 COMPLIANT

In which areas do you use lean or six sigma like Departmental responsibility, Results tracking, Process improvement, Execution of kaizen events etc.

Response: The lean projects are being taken in manufacturing and SCM . Currently 15 lean projects are underway for improvement projects and each project is trac

SECTION I: Regulation Fulfillment and Execution of Studies

Q125 COMPLIANT

Valid Operating Licence?

Response: Yes - factory licence

Q126 COMPLIANT

Environmental License?

Response: Yes - Pollution license

Q127 COMPLIANT

Waste Generation permit?

Response: Yes

Q128 PENDING

Lighting study?

Response: No response recorded

Q129 COMPLIANT

Noise study (plant inside and outside)?

Response: Yes

Q130 COMPLIANT

Atmosphere emissions study?

Response: Yes

Q131 COMPLIANT

Vulnerability study?

Response: N/A

Q132 COMPLIANT

Civil protection approval?

Response: N/A

SECTION J: Energy, Electrical & Utilities (Suggested Additions)

J001 (Suggested Question) PENDING

What is the sanctioned electrical demand, tariff category and voltage level (HT/LT)?

Response: No response recorded

J002 (Suggested Question) — PENDING

What are the connected load, transformer capacities, operating power factor and recorded maximum demand?

Response: No response recorded

J003 (Suggested Question) — PENDING

Are up-to-date electrical single-line diagrams and panel load schedules available?

Response: No response recorded

J004 (Suggested Question) — PENDING

Which main and feeder meters are installed (make, model, accuracy class, CT/PT ratios, calibration date)?

Response: No response recorded

J005 (Suggested Question) — PENDING

How are grid import/export, solar generation, DG generation and energy banking tracked?

Response: No response recorded

J006 (Suggested Question) — PENDING

What are the monthly electricity consumption, energy cost and seasonal demand patterns?

Response: No response recorded

J007 (Suggested Question) — PENDING

Is specific energy consumption (kWh per unit or tonne of product) recorded and benchmarked?

Response: No response recorded

J008 (Suggested Question) — PENDING

Are major machine-wise energy consumption, operating hours, idle energy and peak demand measured?

Response: No response recorded

J009 (Suggested Question) — PENDING

How is power quality monitored (PF, voltage, current imbalance, harmonics, outages and frequency)?

Response: No response recorded

J010 (Suggested Question) — PENDING

What is the status of APFC/capacitor banks, power-factor penalties and reactive power management?

Response: No response recorded

J011 (Suggested Question) — PENDING

How are DG sets tested, maintained and monitored for fuel consumption and kWh output?

Response: No response recorded

J012 (Suggested Question) — PENDING

Are compressed air, HVAC, pumps, motors and other utilities monitored for efficiency and leakage?

Response: No response recorded

J013 (Suggested Question) — PENDING

What is the status of earthing, electrical safety audits, thermography, protection relays and panel labeling?

Response: No response recorded

J014 (Suggested Question) — PENDING

Are there existing energy-saving projects with baseline, estimated savings, capex, owner and verification method?

Response: No response recorded

SECTION K: Digital Systems & Audit Follow-up (Suggested Additions)

K001 (Suggested Question) — PENDING

Which PLC, SCADA, ERP, MES or maintenance systems capture machine and production data?

Response: No response recorded

K002 (Suggested Question) — PENDING

Are machine data ports or communication protocols available (RS-485/Modbus, Ethernet, OPC UA, MQTT)?

Response: No response recorded

K003 (Suggested Question) — PENDING

Who owns operational data and what are its collection intervals, retention and access controls?

Response: No response recorded

K004 (Suggested Question) — PENDING

What is the availability and reliability of site internet, LAN, Wi-Fi and gateway power?

Response: No response recorded

K005 (Suggested Question) — PENDING

How are alerts, abnormal conditions and downtime events acknowledged and resolved?

Response: No response recorded

K006 (Suggested Question) — PENDING

What are the top three improvement opportunities, owners, target dates and expected measurable benefits?

Response: No response recorded

AUDIT SIGN-OFF & REVIEW APPROVAL

Auditor Signature / Submission:

Name: Aarav Sharma (Lead Auditor)

Date: Pending

Reviewer / Lead Auditor Approval:

Name: Dr. Anita Deshmukh (Audit Reviewer)

Date: Pending Review